Benjamin Farsijani

Ottawa, ON | 613-697-1761 | farsijaniben@gmail.com | linkedin.com/in/benfarsi | github.com/benfarsi | benfarsi.com

EDUCATION

University of Ottawa — Honours B.Sc., Computer Science — Sep 2023 – May 2027
GPA: 3.7 / 4.0 — *Ottawa, ON*

Relevant Coursework: Probability & Statistics, Linear Algebra, Calculus, Discrete Mathematics, Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Systems Programming.

TECHNICAL SKILLS

Languages: Python, C, C++, Go, Rust, Bash, SQL

Quantitative & ML: XGBoost, PyTorch, NumPy, pandas, scikit-learn, CasADi, statsmodels; time-series modeling, GARCH(1,1), walk-forward cross-validation, Sharpe / drawdown / hit-rate analysis, Model Predictive Control (MPC), state-space models, stochastic simulation, Monte Carlo methods, constrained optimization

Trading & Market Data: CCXT, Binance API, REST/WebSocket market data, order-book and OHLCV processing, backtesting, risk management (TP/SL, fee modeling, position sizing)

Systems & Performance: Linux (POSIX), multithreading, lock-free concurrency, low-latency networking, TCP/IP, epoll, non-blocking I/O

Infrastructure & Tools: PostgreSQL, SQLite, Docker, Git, GDB, Flask

PROJECTS

Benny — Algorithmic Crypto Trading System | *Python, XGBoost, CCXT, Binance API, PostgreSQL, Docker*

- Built end-to-end ML trading pipeline ingesting real-time market data from Binance across 6 liquid crypto assets (BTC, ETH, SOL, BNB, XRP, DOGE); engineered 50+ features including GARCH(1,1) conditional volatility, Parkinson high-low range estimator, VWAP deviation, OBV, ADX, Stochastic RSI, and multi-timeframe log returns.
- Trained XGBoost classifier for directional price prediction with strict zero-lookahead feature construction; validated using 5-fold walk-forward cross-validation, reporting per-fold Sharpe ratio, max drawdown, hit rate, and simulated P&L versus buy-and-hold baseline.
- Designed full risk-management layer: probability-gated signal confidence (long if p > 0.65, exit if p < 0.35), 3% take-profit / 1.5% stop-loss thresholds, 0.1% maker/taker fee modeling, and per-symbol position sizing across a $6,000 paper portfolio.
- Implemented RSI + EMA fallback strategy and ran parametric threshold sweep over (RSI period, EMA spread, entry/exit cutoffs); benchmarked annualized Sharpe, max drawdown, win rate, and turnover against buy-and-hold to identify regime-robust parameter sets.
- Deployed live containerized dashboard (Flask, Chart.js, Docker, PostgreSQL) tracking equity curve, per-symbol P&L, signal state, and trade history; system runs continuously against live Binance market data.

Stochastic Control & MPC for Thermal Oil Recovery | *Python, PyTorch, NumPy, CasADi*

- Developed stochastic simulation environment for thermal oil extraction modeling delayed nonlinear dynamics between steam injection, reservoir temperature, and production output; generated 100k+ synthetic multivariate time-series observations for control experimentation.
- Implemented constrained Model Predictive Control (MPC) and learned state-space forecasting models in PyTorch and CasADi to optimize steam injection policies, targeting a projected 5–10% reduction in simulated steam-to-oil ratio (SOR) versus heuristic baselines.
- Researched partially observed dynamical systems, stochastic optimization, and sequential decision-making methods for adaptive industrial process control, with applications to energy systems, quantitative modeling, and uncertainty-aware optimization.

EXPERIENCE

Technical Mentor — Embedded Systems & Prototyping — Sep 2024 – Present
Centre for Entrepreneurship and Engineering Design (CEED), University of Ottawa — *Ottawa, ON*

- Mentored engineering students through firmware debugging, peripheral integration (I2C, SPI, UART), and hardware bring-up in CEED's Makerspace; supported projects involving microcontrollers, sensors, and rapid-prototyping equipment.
- Guided hardware/software co-design decisions across multiple student teams, accelerating prototype iteration cycles and resolving embedded integration issues under tight competition deadlines.